

Acute alkalinity stress triggers kidney histopathology, oxidative-immune dysregulation, and MAPK/TGF- β signaling pathways activation in *Scatophagus argus*

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ABSTRACT

Saline-alkaline water aquaculture offers a promising approach to address water scarcity and increase aquatic product yields. This study investigates the effects of acute alkalinity stress on spotted scat (*Scatophagus argus*), a species known for its adaptability to various salinity conditions. We exposed *S. argus* to alkalinity stress for different durations (0, 12, 24, 48, 72, and 96 h) under two experimental conditions: a control group (C) with 30 ‰ salinity and 0 mmol/L alkalinity, and a treatment group (T) with 30 ‰ salinity and 30 mmol/L alkalinity. Following stress exposure, we conducted histological, biochemical, and transcriptomic analyses to assess the impact on kidney tissue. Histological results showed significant kidney damage resulting from acute alkalinity stress, characterized by enlarged nuclei, vacuolization, and necrosis of renal tubules and glomeruli. Biochemical assays revealed altered enzyme activities: levels of superoxide dismutase (SOD) and catalase (CAT) decreased, with SOD exhibiting a slight rebound at 72 h before declining again, indicating a disrupted redox homeostasis. In contrast, the activities of glutathione peroxidase (GSH-Px), malondialdehyde (MDA), acid phosphatase (ACP), and alkaline phosphatase (ALP) increased over time, reflecting oxidative stress and disruptions in metabolic and immune functions. Transcriptomic analysis identified 247 differentially expressed genes (DEGs), with 109 up-regulated and 138 down-regulated. These DEGs were significantly enriched in pathways related to immune response, cell adhesion, and protein folding, with the MAPK and TGF- β signaling pathways playing crucial roles in the stress response. These results suggest that acute alkalinity stress induces kidney damage, disrupts immune and oxidative homeostasis, and alters gene expression in *S. argus*, providing a deeper understanding of the molecular mechanisms underlying stress responses in saline-alkaline conditions.

1. Introduction

Since the beginning of the 21st century, the global aquaculture industry has experienced rapid development, consistently increasing its contribution to global fish production. It is estimated that by 2032, global aquatic animal production will reach 205 million tons, with aquaculture's share rising from the current 51 % (Naylor et al., 2021; FAO, 2024). United Nations' World Population Prospects 2019 anticipates that the global population will reach 9.7 billion by 2050, thereby

intensifying the pressure on food production systems. Meeting the consumption needs of this growing population by producing more high-quality and safe aquatic products will be a significant challenge. Given that most natural fishery resources are already fully exploited or over-exploited, further expansion of aquaculture has become essential (United Nations, 2019; Partelow et al., 2025). Saline-alkaline water aquaculture, as an emerging farming model, offers a solution to the growing issue of water resource shortages in traditional aquaculture. It has the potential to enhance the supply of aquatic products while

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delivering economic, social, and ecological benefits (Anufrieva, 2018). However, studies indicate that saline-alkaline water aquaculture can disrupt the homeostasis of fish, adversely affecting osmoregulation, growth, and development, and in severe cases, leading to mortality (Chang and Liang, 2021; Li et al., 2024a, 2024b). Moreover, the immune system characteristics and stress resistance mechanisms in cultured species under these conditions remain poorly understood (Zhou et al., 2024a; Zhou et al., 2024b).

Alkalinity, a critical environmental factor in saline-alkaline waters, directly impacts the physiological state of aquatic animals (Qin et al., 2021). Previous studies have showed that alkalinity act as a buffering agent in environment, optimal alkalinity levels can enhance the resilience of aquatic species to environmental stress (Li et al., 2024a, 2024b). However, excessive alkalinity can disrupt the osmotic balance and ion homeostasis of fish body fluids, negatively impacting their growth, antioxidant capacity, and immune function (Pang et al., 2021). Wilkie and Wood (1996) suggested that high alkalinity primarily affects fish by disrupting their osmotic balance and ion exchange systems in the gills and cell membranes. This disturbance compromises the acid-base buffering system, potentially leading to mortality. Similarly, research on the lahontan cutthroat trout (*Oncorhynchus clarkii henshawi*) has demonstrated that under alkaline conditions, the fish's metabolic rate increases, resulting in elevated internal ammonia nitrogen levels. This rise in ammonia was identified as the primary cause of mass mortality due to ammonia toxicity (Wright et al., 1993). Although changes in alkalinity can disrupt the normal biological processes, fish have evolved the ability to adapt through active regulatory and metabolic mechanisms, allowing them to maintain normal physiological function within specific salinity and alkalinity ranges (Wen et al., 2023). For instance, Songpu carp (*Cyprinus carpio* Songpu) demonstrates exceptional adaptability to highly alkaline environments through precise regulation of ion transport in its gills and enhanced renal functions, which accelerate the clearance of excess hydroxide ions (OH^-) and bicarbonate (HCO_3^-), reducing the risk of systemic alkalosis (Fan et al., 2021). Similarly, naked carp (*Gymnocypris przewalskii*), another Cypriniformes species, maintains effective gas exchange and osmotic pressure balance in highly alkaline environments through tightly arranged renal tubules and long, densely packed gill rakers (Tong et al., 2015; Zhou et al., 2025). Other species, such as *alburnus tarichi* (*Oxyaemacheilus ercisanus*) (Akku et al., 2021) and amur ide (*Leuciscus waleckii*) (Xu et al., 2013a, 2013b), which have long inhabited extreme alkaline environments, regulate physiological processes such as osmotic pressure, immune defense, and ammonia nitrogen excretion to thrive in saline-alkaline conditions. These findings highlight the distinct stress responses and adaptation mechanisms of different fish species to alkalinity stress, underscoring the need for further research in this area. However, current studies have primarily focused on immune adaptation or immunosuppression in freshwater fish under high alkalinity, with a significant gap remaining in research on the adaptation mechanisms of marine fish to acute alkalinity stress.

The kidney plays a crucial role in the stress metabolism response of fish, regulating hormone secretion, metabolic detoxification, immune function, and other adaptive processes (Petitjean et al., 2019). However, it is also a highly sensitive organ, where even brief exposure to alkalinity stress can alter antioxidant capacity and cause minor tissue damage (Kao et al., 2010). Thus, the kidney serves as a reliable indicator of fish responses to alkalinity stress. Numerous studies have shown that acute alkalinity stress affects kidney function, physiological and biochemical parameters, and the expression of metabolic and immune-related pathways in fish. For example, Ding et al. (2023) found that acute alkalinity stress in crucian carp (*Carassius auratus*) significantly disrupts metabolic pathways involved in energy allocation, enhances antioxidant enzyme activity to mitigate oxidative stress, and regulates blood urea nitrogen and creatinine levels, reflecting both an adaptive response and renal repair capacity. Similarly, Kültz (2015) reported that saline-alkaline stress in *C. auratus* disrupts antioxidant systems and energy homeostasis, inhibits protein and amino acid metabolism, and leads to purine and

lipid metabolism disorders, potentially causing severe kidney damage. Oguz (2015) observed that in saline-alkali lakes, van fish (*Alburnus tarichi*) exhibited significantly lower $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity in the kidney compared to freshwater environments, alongside structural collapse of glomeruli and narrowing of collecting tubules. Furthermore, Wang et al. (2003) demonstrated that freshwater *L. waleckii* adapted to high carbonate alkalinity by upregulating kidney phosphatase gene expression. These findings underscore the dual role of the kidney as a critical regulatory organ facilitating metabolic and structural adaptation, and as a vulnerable target susceptible to damage under prolonged or severe stress. This highlights the need for further research to deepen our understanding of renal adaptation mechanisms and develop strategies to mitigate the impacts of alkalinity stress on fish health.

The Perciformes are extensively studied across various biogeographic scales, ranging from regional to global levels (Fahay, 2007). Relevant research has examined the effects of environmental factors, such as salinity and alkalinity, on growth, reproduction, nutrition, immunity, and other biological aspects (Li et al., 2024a, 2024b; Zhao et al., 2020; Wang et al., 2025; Yao et al., 2010). Among these species, the spotted scat (*Scatophagus argus*), a member of the order Perciformes, is commonly found in estuaries and mangrove coastal areas of the Indo-Pacific region. Due to its tender meat, high adaptability, and strong disease resistance, *S. argus* is considered one of the preferred economic species for aquaculture (Gupta, 2016; Barry and Fast, 1992; Sivan, 2011; Gandhi, 2002). As a euryhaline fish, *S. argus* can thrive in waters with varying salinities, inhabiting both seawater and freshwater environments, and demonstrates optimal growth at a pH range of 5.0 to 9.0. Despite its adaptability and resilience, several challenges remain in farming this species in saline-alkaline waters (Afzal et al., 2023; Wang et al., 2024). Previous studies have demonstrated that the osmoregulatory organs of *S. argus*, such as the gills and kidneys, undergo adaptive modifications to cope with fluctuations in environmental salinity (Mu et al., 2015). Building on this foundational understanding, recent research employing RNA-Seq technology has revealed that salinity stress significantly affects liver lipid metabolism in *S. argus* by differentially regulating lipid metabolism-related genes under both low- and high-salinity conditions, highlighting their critical role in the fish's salinity adaptation (Chen et al., 2023). However, while various fish species have been shown to counteract alkalinity stress through adaptive organ modifications, specialized ion transport, and metabolic adjustments, the specific mechanisms employed by the marine species *S. argus* remain poorly understood. Addressing this gap is crucial for gaining a comprehensive understanding of *S. argus* alkalinity tolerance and could provide valuable insights for advancing aquaculture practices in saline-alkaline environments.

This study aims to investigate the response of *S. argus* to acute alkalinity stress by examining histological changes, enzyme activity, and transcriptomic alterations. The findings will provide insights into the high alkaline adaptability of *S. argus* and contribute to the development of aquaculture strategies for saline-alkaline water systems.

2. Materials and methods

2.1. Ethics statement

All experimental protocols in this study adhered to the standards set by the Animal Research and Ethics Committee of Guangdong Ocean University (NIH Pub. No.85–23, revised 1996). The species involved are neither endangered nor protected.

2.2. Animals and sample collection

Juvenile *S. argus* were sourced from the Donghai Island Experimental Base in Zhanjiang City, Guangdong Province, China. Following the methodology described by Overton et al. (2008), experimental water was prepared by mixing seawater and freshwater to achieve the desired

salinity. Analytical-grade NaHCO_3 (Aladdin, Shanghai) was added to adjust alkalinity (Qin et al., 2021; Zhang et al., 2023c). Prior to the experiment, fish were acclimated in 60 L tank for two weeks. During the experiment, water temperature was maintained at $32 \pm 1^\circ\text{C}$, dissolved oxygen at 6.4 ± 0.2 mg/L, pH between 8.0 and 8.3, and ammonia nitrogen below 1 mg/L. Fish were randomly divided into two groups: the control (C) group and the treatment (T) group, with three replicates per group. Experimental conditions were determined based on results from a preliminary alkalinity tolerance trial, in which fish were exposed to a range of alkalinity levels and mortality was recorded at 24, 48, and 96 h. Based on the calculated LC_{50} , the treatment group (T) was exposed to a salinity of 30 ‰ and alkalinity of 30 mmol/L, while the control group (C) was maintained at 30 ‰ salinity with no additional alkalinity (Fig. 1 A-B). The fish in each group had an average body weight of 5.65 ± 1.16 g and an average body length of 4.46 ± 0.34 cm. After reaching the final concentration, kidney tissue samples were collected from six individuals per group at each of the following time points: 0, 12, 24, 48, 72, and 96 h. One portion of the kidney tissue was stored in liquid nitrogen at -80°C , while another portion was preserved in Bouin's liquid for subsequent experiments. The experimental process is detailed in Fig. 1C.

2.3. Histological analysis of kidney

Histological analysis of kidney tissues was performed on samples randomly selected from six individuals in each group. After fixation in Bouin's solution for 24 h, the kidney samples underwent gradient dehydration using ethanol, followed by embedding in paraffin and

sectioning. Subsequently, histopathological changes in the kidney were visualized and documented under a microscope following hematoxylin-eosin (HE) staining.

2.4. Kidney biochemical indices

Kidney concentrations of key enzymes including catalase (CAT), superoxide dismutase (SOD), and malonaldehyde (MDA), were measured in six randomly selected individuals from each group using an Enzyme-Linked Immunosorbent Assay (ELISA) kit (Sinobestbio, Shanghai) according to the manufacturer's instructions. To reduce the impact of individual variability, samples were pooled in pairs, and each group was analyzed with three biological replicates, each consisting of two technical replicates.

2.5. RNA extraction, library construction, sequencing and assembly

Total RNA was extracted from kidney tissues of both the control and treatment groups (six samples per group) at 24-h using Trizol reagent (Invitrogen, Waltham, MA, USA), following the manufacturer's protocol from Takara (Shiga, Japan). RNA quality was assessed using the TRIzol Reagent Kit and confirmed by 1 % agarose gel electrophoresis. RNA integrity and quantity were further evaluated using the Agilent Bioanalyzer 2100 system (Wang et al., 2023). To reduce the effects of individual variability, samples were pooled in pairs. A total of six RNA libraries were established using the NEBNext® Ultra™ RNA Library Prep Kit (NEB, USA), guided by the manufacturer's protocol. After library construction, the concentration was quantified using a Qubit 2.0

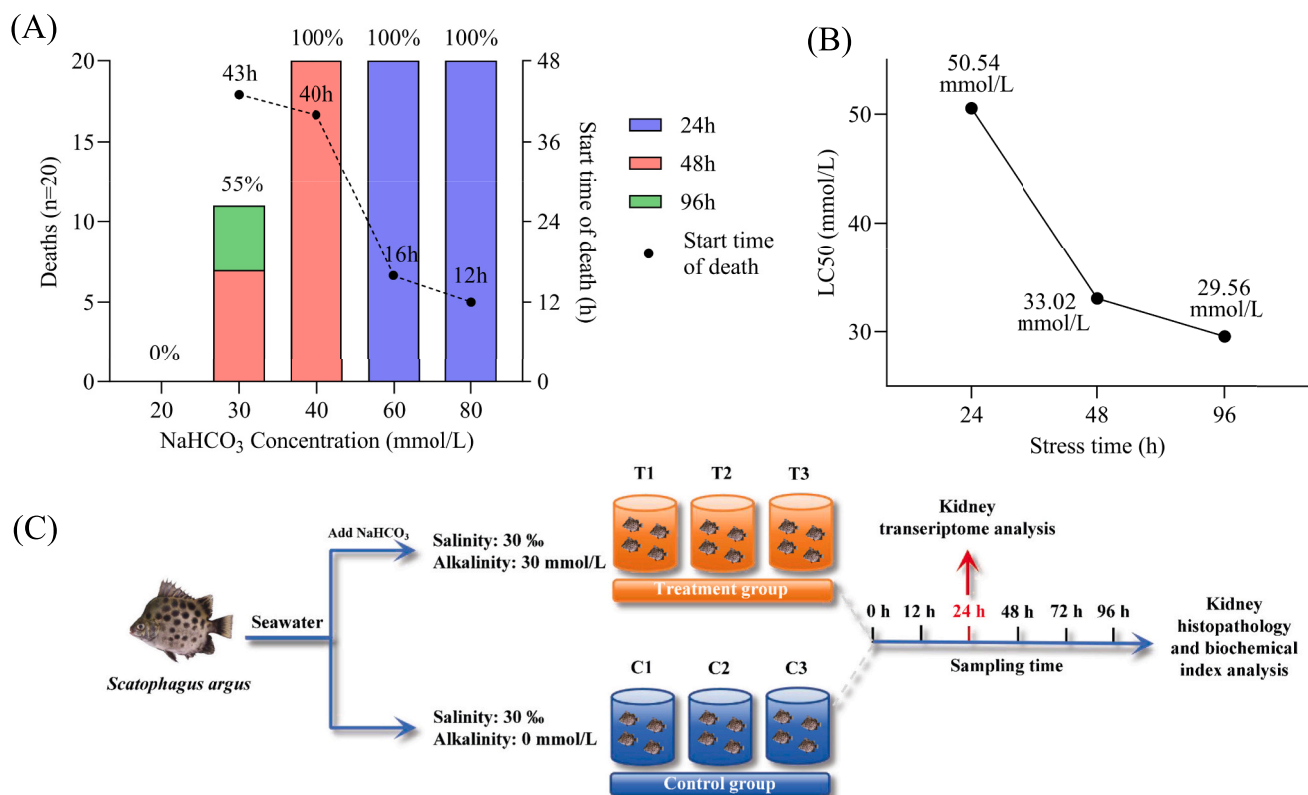


Fig. 1. (A) Deaths and Mortality statistics of *S. argus* under different NaHCO_3 concentrations at 24 h, 48 h, and 96 h. Blue means number of deaths at 24 h, red means number of deaths at 48 h, green means number of deaths at 96 h, percentage represents mortality, and dotted line represents the start time of death. (B) semi-lethal concentration (LC_{50}) statistics of *S. argus* under different NaHCO_3 concentrations at 24 h, 48 h, and 96 h. (C) The experimental design diagram of acute alkalinity stress in *S. argus*. Three replicates were selected from the kidney tissue of the control group, labeled as Control 1–3 (C1, C2, C3). Similarly, three replicates were chosen from the 24-h alkalinity stress treatment group, labeled as Treatment 1–3 (T1, T2, T3). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fluorometer and subsequently diluted to 1.5 ng/μl. The insert size of the library was assessed using the Agilent 2100 Bioanalyzer. Subsequently, the effective concentration of the library was determined by qRT-PCR, ensuring it exceeded 2 nM for optimal quality. Sequencing was performed using the Illumina NovaSeq™ 6000 (Illumina, USA), generating 150 bp paired-end reads. Low-quality reads in the raw fastq data were removed, and the filtered reads were assembled using Trinity software (Garber et al., 2011) with default parameters to obtain clean reads. The paired-end clean reads were then mapped to the reference genome (Huang et al., 2021) using HISAT2 v 2.0.5. RNA-Seq data are available in NCBI's SRA database under the accession number PRJNA1187670.

2.6. Differentially expressed genes (DEGs) and functional annotation

Clean reads were aligned to the reference genome using Hisat2. Gene expression levels were quantified using the featureCounts tool (version 1.5.0-p3), which counts the number of reads mapping to each gene. The Fragments Per Kilobase of transcript per million mapped reads (FPKM) values were calculated, accounting for gene length (Liao et al., 2014). DEGs for each comparison group were identified using the DESeq2 package, with multiple comparisons corrected using the Benjamini-Hochberg false discovery rate (FDR) method. Significant DEGs selected based on an adjusted p -value (padj) < 0.05 and an absolute $\log_2$ fold change ($\log_2\text{FC}$) > 1 (Anders and Huber, 2010). To annotate the functional roles of these genes, the sequences were compared against zebrafish (*Danio rerio*) in NCBI databases to predict their functions and potential involvement in various biological processes. To gain further insights into the biological significance of the identified DEGs, the clusterProfiler package was used for Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses (Yu et al., 2012).

2.7. Validation of gene expression in enrichment pathways by qRT-PCR

To ensure the accuracy of the sequencing results and validate gene expression in enriched KEGG signaling pathways, quantitative reverse transcription polymerase chain reaction (qRT-PCR) was conducted on 10 significantly DEGs. Specific primers were designed using Primer 6.0 and synthesized by Sangon Biotechnology Co., Ltd. (Shanghai, China), as outlined in Table S1. Total RNA was extracted from kidney samples (a total of 6 samples) from both groups using the TRIzol reagent kit. The RNA was then reverse-transcribed into complementary DNA (cDNA) using the TransScript® Fly First-Strand cDNA Synthesis SuperMix kit (Beijing TransGen Biotech Co. Ltd., Beijing, China), following the manufacturer's instructions. qRT-PCR was performed using the PerfectStart® Green qPCR SuperMix (Beijing TransGen Biotech Co., Ltd., Beijing, China) in a Roche LightCycler 384 real-time PCR system. The reaction protocol included an initial pre-denaturation at 95 °C for 5 min, followed by 40 cycles of 95 °C for 30 s and 52 °C for 30 s. β -actin served as the internal reference for gene expression normalization, and fold change in target gene expression was calculated using the $2^{-\Delta\Delta\text{Ct}}$ method.

2.8. Statistical analysis

Statistical analyses were conducted using SPSS Statistics (version 26) to ensure accuracy and reliability. Data were presented as means $\pm$ standard deviations (SD) for variables following normal distributions. To determine the statistical significance of differences in mean values between the C and T groups, a one-way ANOVA was utilized. A p -value < 0.05 was considered statistically significant, while p -values < 0.01 indicated highly significant differences.

3. Result

3.1. Alkalinity tolerance of *S. argus*

Alkalinity tolerance results indicated that *S. argus* exhibits strong adaptability under low alkaline conditions (20 mmol/L), with no deaths observed throughout the experiment. However, as NaHCO₃ concentration increased, the species' resistance to alkaline stress diminished rapidly. At 30 mmol/L, no significant mortality was recorded at 24 h; however, by 96 h, mortality had reached 55 %, suggesting that even moderate alkalinity can induce toxic effects over time. At higher concentrations (40 mmol/L and above), *S. argus* showed markedly reduced resilience. Exposure to 40 mmol/L led to 100 % mortality by 48 h. At concentrations of 60 mmol/L and 80 mmol/L, fatalities occurred as early as 16 and 12 h, respectively, with complete mortality within 24 h. These results clearly demonstrate that elevated NaHCO₃ levels rapidly produce lethal effects on *S. argus*. Furthermore, the progressively declining LC₅₀ over time further confirms the cumulative, time-dependent impact of alkalinity on the survival of this species (Fig. 1A-B).

3.2. Effects of acute alkalinity stress on kidney tissue structure

Histological analysis of kidney tissue using HE staining revealed that prior to acute alkalinity stress, kidney cells were arranged in an orderly manner with distinct boundaries. As the stress progressed, the nuclei of the kidney cells significantly enlarged, leading to a disordered arrangement, an increase in gaps, and the emergence of vacuolar degeneration in some cells, indicative of inflammatory infiltration. After 72 h of alkalinity stress, some renal tubular epithelial cells exhibited injury, characterized by a noticeable reduction in both the size and number of nuclei. Additionally, phenomena such as glomerular shrinkage, renal tubular degeneration, tubular separation, and tubular necrosis began to manifest, becoming increasingly pronounced with the duration of the stress (Fig. 2).

3.3. Effects of acute alkalinity stress on renal biochemical indices

The changes in biochemical indicators of kidney function at different treatment times indicated that under acute alkalinity stress, the activities of superoxide dismutase (SOD) and catalase (CAT) decreased, with SOD exhibiting a slight rebound at 72 h before declining again. Conversely, the activities of glutathione peroxidase (GSH-Px), malondialdehyde (MDA), acid phosphatase (ACP), and alkaline phosphatase (ALP) demonstrated an upward trend as treatment time extended, with ACP, ALP, and MDA reaching their peak levels at 96 h. Significance tests showed that the activities of SOD and CAT in the control (C) group were significantly lower than those in the treatment (T) group, while the activities of GSH-Px, ACP, ALP, and MDA were significantly higher in the T group compared to the C group ($P < 0.05$; Fig. 3).

3.4. Transcriptome sequencing analysis

3.4.1. Sequencing of reads

Six kidney tissue samples (three each from the control and treatment groups) of *S. argus* exposed to 24 h of acute alkalinity stress were sequenced, generating an average of 44.51 million clean reads per sample. The percentage of Q30 bases exceeded 92.58 %, and the GC content ranged from 48.48 % to 49.15 %. These clean reads were then aligned with the genome sequence of *S. argus*, with a comparison rate of over 87.42 %. These results indicate that the sequencing data generated in this experiment is of high quality and meets the standard for subsequent analysis (Table 1).

3.4.2. Differential expression analysis

Under acute alkalinity stress, a total of 247 DEGs were identified. Among them, 109 genes were up-regulated, and 138 genes were down-

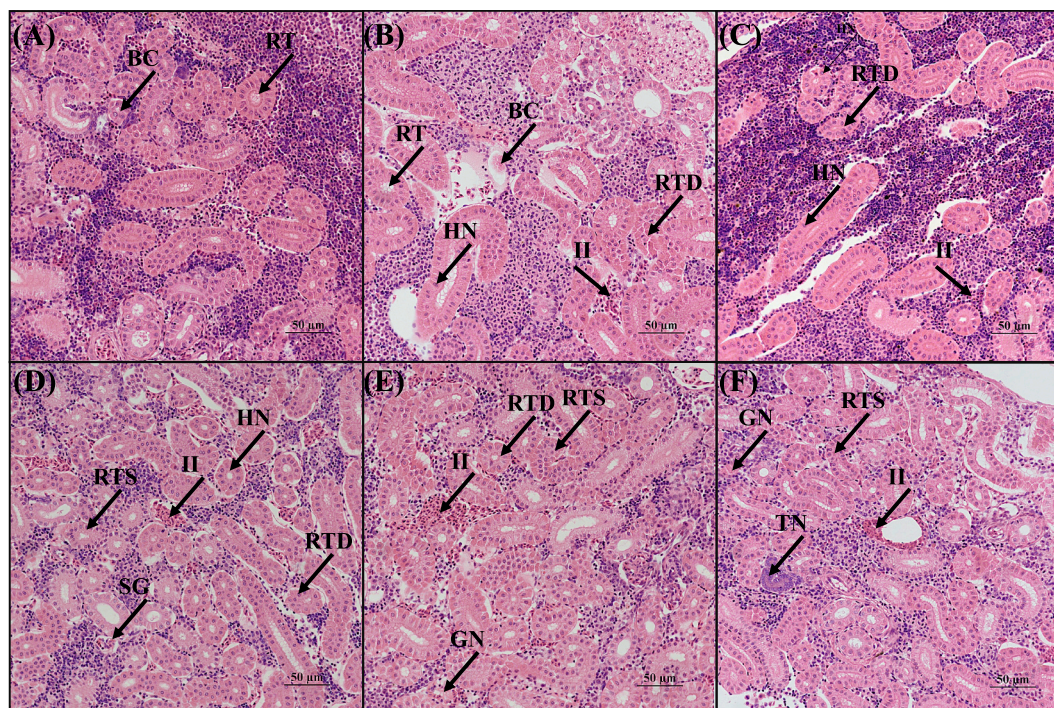


Fig. 2. Kidney structure of *S. argus* at different treatment stages. (A): 0 h control group, (B): 12 h treatment group, (C): 24 h treatment group, (D): 48 h treatment group, (E): 72 h treatment group, (F): 96 h treatment group. HN: Pathological cell nucleus, II: Inflammatory infiltration, RT: Normal appearance of renal tubules, BC: Bowman's capsule, GN: Glomerular necrosis, SG: Shrinking of the glomeruli, RTD: Renal tubular degeneration, RTS: Renal tubular separation, TN: Tubular necrosis. Scale bar: 50 μm .

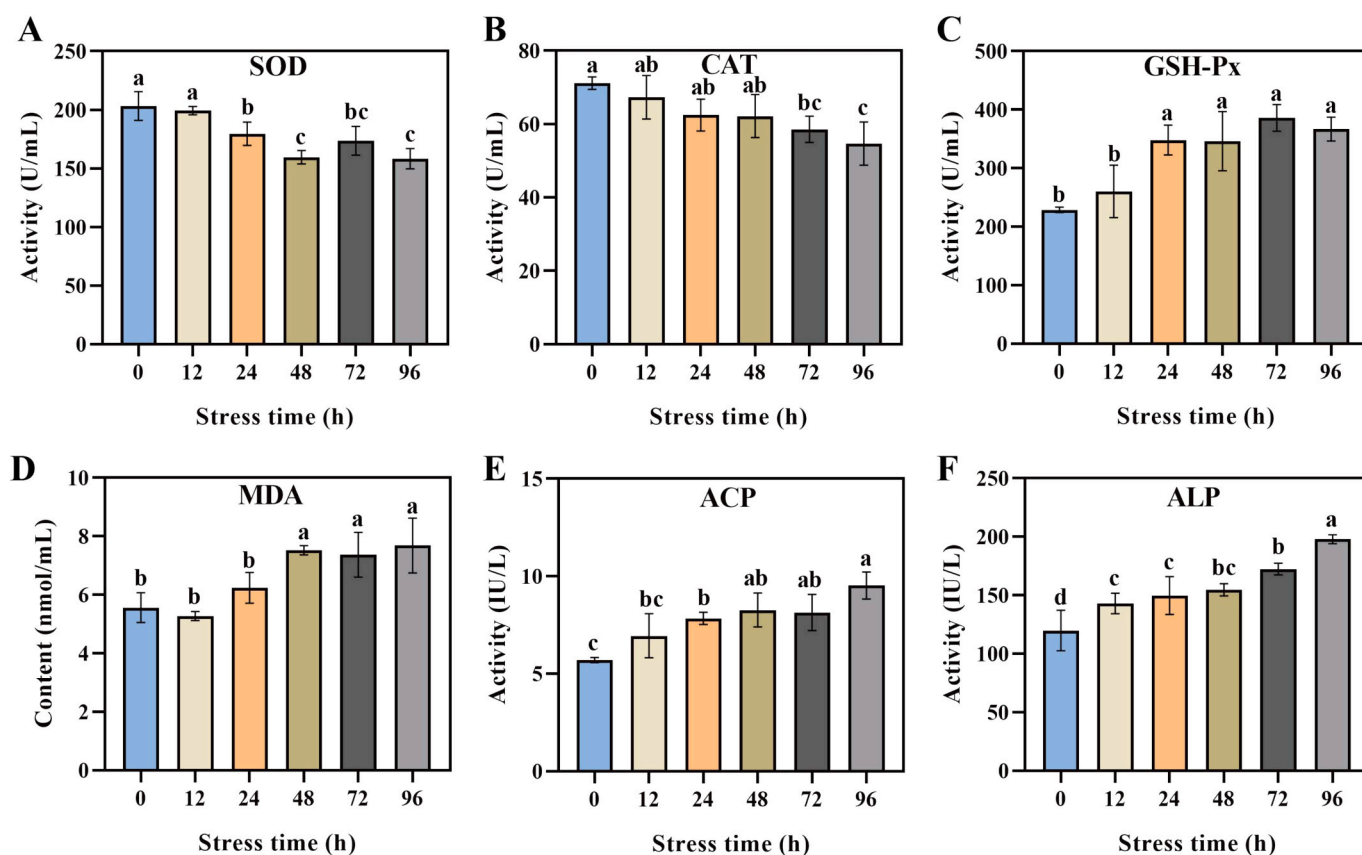


Fig. 3. Biochemical indexes levels in the kidney at different treatment times. Data are presented as mean $\pm$ SD ($n = 6$). Means with different letters at the same point indicate significant differences between treatment times ($P < 0.05$), while means with the same letter indicate no significant difference ($P > 0.05$).

Table 1
Quality metrics of kidney tissue sequencing data from *S. argus* under acute alkalinity stress.

Sample	Raw_reads	Raw_bases (G)	Clean_reads	Clean_bases (G)	Error_rate	Q20 (%)	Q30 (%)	GC_pct
C1-ki	47,057,694	7.06	44,981,610	6.75	0.03	97.32	92.88	48.98
C2-ki	43,635,698	6.55	413,58,568	6.20	0.03	97.14	92.47	49.00
C3-ki	49,312,628	7.40	48,134,252	7.22	0.03	97.52	93.35	49.24
T1-ki	40,092,948	6.01	38,349,198	5.75	0.03	97.45	93.14	48.48
T2-ki	43,868,744	6.58	42,135,426	6.32	0.03	97.26	92.69	49.15
T3-ki	48,169,172	7.23	46,690,202	7.00	0.02	98.11	94.6	48.93

Note: Three replicates were selected from the kidney tissue of the control group, labeled as Control 1–3 (C1, C2, C3). Similarly, three replicates were chosen from the 24-h alkalinity stress treatment group, labeled as Treatment 1–3 (T1, T2, T3).

regulated in the kidney tissue (Fig. 4A). Furthermore, the heatmap of DEGs expression patterns demonstrated consistent expression trends for individual genes and clear clustering relationships between groups in the kidney tissue under acute alkalinity stress (Fig. 4B).

3.4.3. Functional classification of DEGs

To further understand the biological significance of DEGs in kidney tissue under acute alkalinity stress, GO enrichment analysis was performed. The results revealed significant enrichment of DEGs in biological processes (BP), cellular components (CC), and molecular functions (MF). In the BP category, DEGs were notably enriched in immune response, immune system processes, cell adhesion, biological adhesion, protein folding, and other aspects. The CC category showed significant enrichment in the troponin complex, striated muscle, and thin filament myofibrils. In the MF category, DEGs were significantly enriched in cytokine activity, protein dimerization activity, unfolded protein binding, and cytokine receptor binding ($P < 0.05$; Fig. 5A).

Furthermore, KEGG pathway analysis identified 29 genes

significantly enriched in 13 pathways, including the MAPK signaling pathway, motor proteins, TGF- β signaling pathway, and phagosome-related pathways ($P < 0.05$). Notably, DEGs such as myc proto-oncogene beta (*myc-b*), transforming growth factor beta 3 (*tgfb3*), udp-glucuronosyltransferase 3 (*ugt3*), tubulin beta class i (*tubb*), tubulin alpha 1c (*tuba1c*), and troponin t3 fast skeletal type (*tnnt3d*) were present in multiple significantly enriched pathways (Fig. 5B).

3.4.4. Interaction between MAPK and TGF- β Pathways

To further investigate the mechanisms underlying alkalinity stress response in *S. argus*, we analyzed the expression profiles of DEGs within these 13 pathways (Fig. 6A–C). Gene expression analysis revealed that under alkalinity stress, 21 DEGs were down-regulated and 29 DEGs were up-regulated (Fig. 6A). Notably, the majority of DEGs in the four most significantly enriched pathways were down-regulated (Fig. 6B). KEGG enrichment analysis indicated that the MAPK and TGF- β pathways showed the highest DEGs enrichment, suggesting they play a central role in the alkalinity stress response. In the MAPK pathway, protein

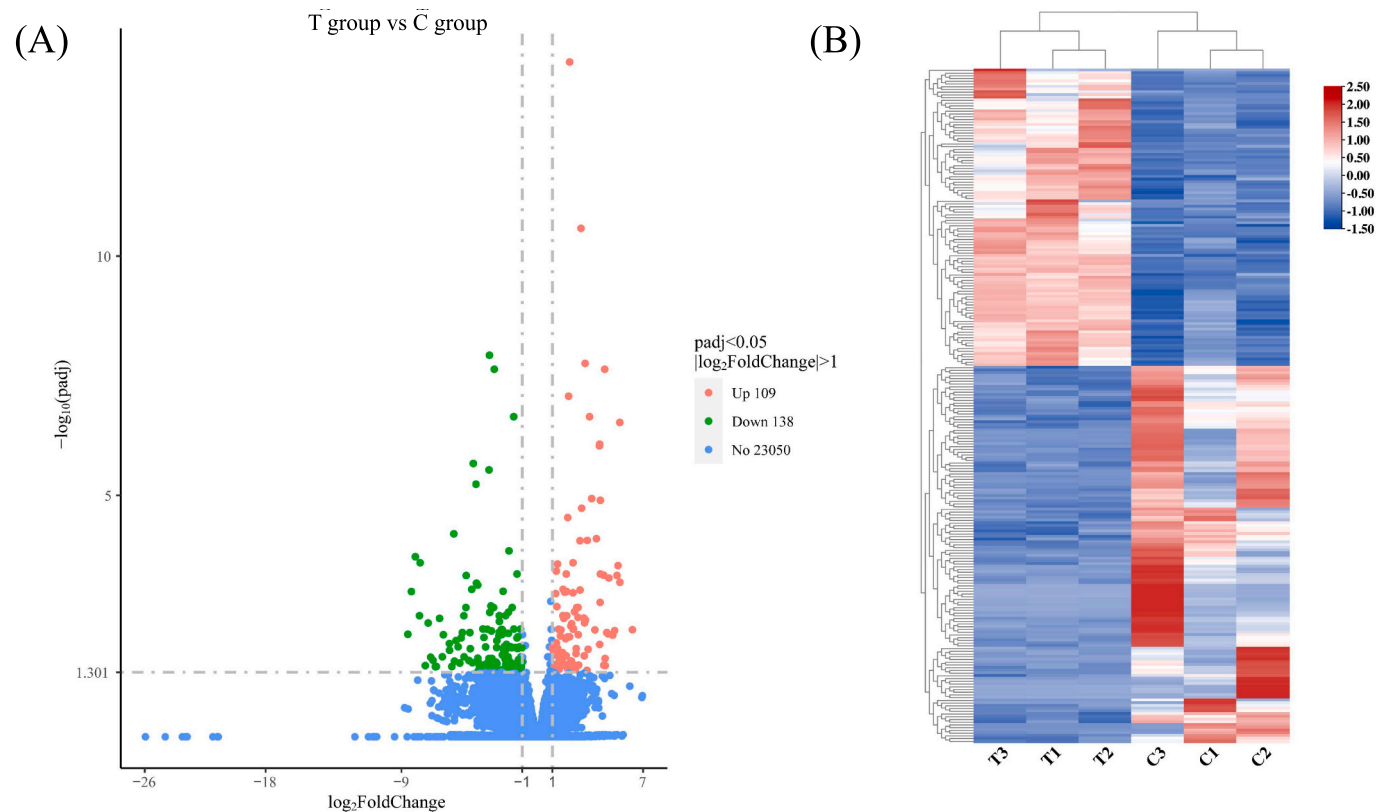


Fig. 4. (A) The number of DEGs between the C and T (24 h) groups. The x-axis represents the fold change in gene expression, and the y-axis represents the significance of DEGs. Red, green and blue represent up-regulated, down-regulated and normal DEGs, respectively. (B) Heatmap analysis of hierarchical clustering for all DEGs in the C group and T group. The x-axis represents the C and T groups, and the y-axis represents each DEG. The $\log_2$ (FPKM + 1) values were normalized and transformed (scale number) before clustering. Red indicated high gene expression and blue indicated low gene expression. The colour is from red to blue, indicating that $\log_2$ (FPKM + 1) is from large to small. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

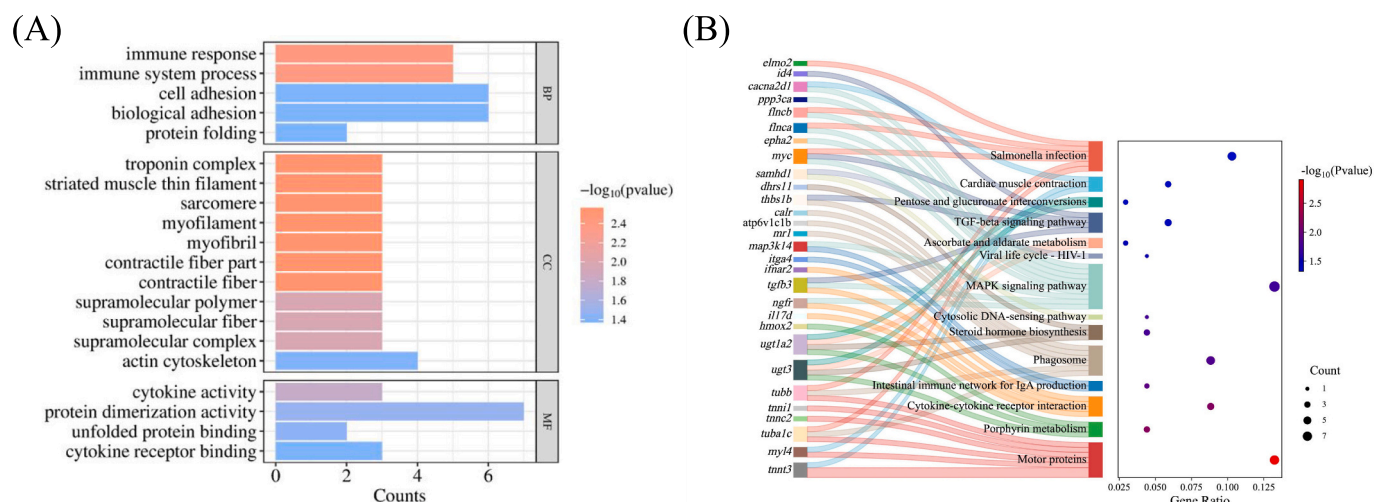


Fig. 5. (A) GO enrichment column diagram of DEGs in T group vs C group. The abscissa is the number of DEGs annotated to the GO pathway, and the ordinate is the GO pathway. (B) KEGG enrichment bubble diagram of DEGs in T group vs C group. The abscissa is the ratio of the number of DEGs annotated to the KEGG pathway to the total number of DEGs, and the ordinate is the KEGG pathway.

phosphatase 3 catalytic subunit alpha (*ppp3ca*) was up-regulated. Conversely, several genes, including calcium voltage-gated channel auxiliary subunit alpha2 delta 1 (*cacna2d1*), *ngfr* antisense alpha (*ngfra*), eph receptor a2 beta (*epha2b*), *tgfb3*, filamin c alpha (*flnca*), filamin c beta (*flncb*), and *myc-b*, were down-regulated. In the TGF- β pathway, the inhibitor of dna binding 4 (*id4*) was up-regulated, while genes including thrombospondin 1 (*ths1b*), *tgfb3*, and *myc-b*, were down-regulated.

3.5. qRT-PCR validation of transcriptome data

Based on the results of KEGG enrichment analysis, 10 DEGs from MAPK and TGF- β signaling pathways were selected for qRT-PCR verification: *cacna2d1*, *ngfra*, *epha2b*, thrombospondin-1 (*ths1b*), *tgfb3*, *flnca*, *flncb*, *ppp3ca*, *id4*, *myc-b*. The expression patterns of all selected genes were consistent between RNA-seq and qRT-PCR (Fig. 7), further confirming the accuracy and reliability of the RNA-seq data and validating the gene expression profiles of MAPK and TGF- β signaling pathways.

4. Discussion

Alkalinity is a critical external factor influencing fish survival (Smyth and Elliott, 2016). Drastic fluctuations in alkalinity can trigger stress responses in fish, leading to endocrine disorder, immune system damage, and impaired growth and development, which in severe cases can threaten their survival (Albecker and McCoy, 2017). This study investigated the response and adaptability of *S. argus* to acute alkalinity stress. We observed histological changes in the kidney after acute alkalinity stress and compared the enzyme activity and transcriptome. Our results indicated that acute alkalinity stress caused kidney cells damage and altered the activity of enzymes involved in immune and metabolic pathways. Transcriptome analysis further supported these findings, providing insight into the molecular mechanisms underlying the stress response of *S. argus* to acute alkalinity stress.

4.1. Acute alkaline treatment induces kidney histopathological damage

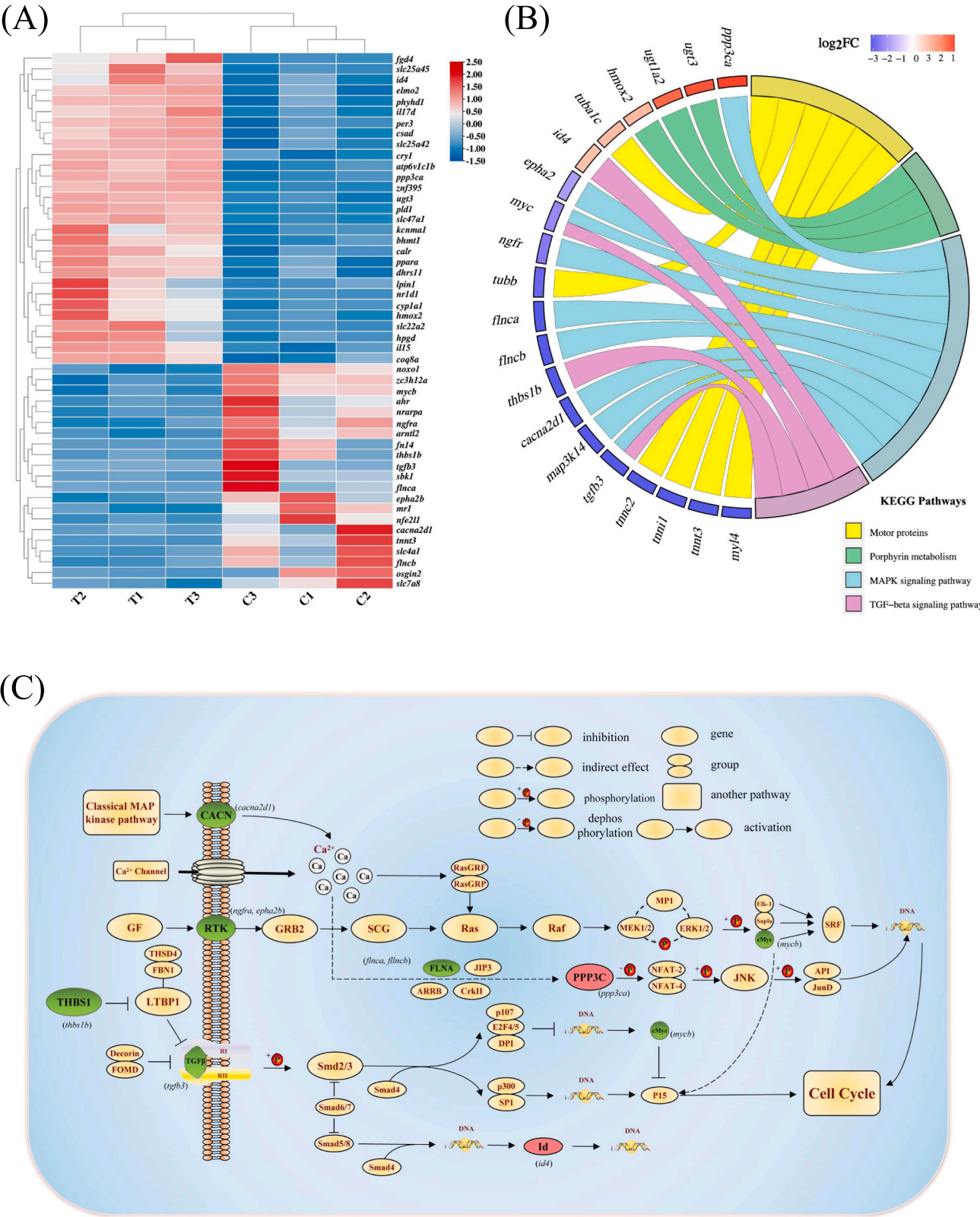
Numerous studies have demonstrated that exposure to high alkalinity stress induces a range of physiological changes and tissue damage in fish (Fang et al., 2022; Menon et al., 2023; Wang et al., 2024). Traditionally, it was believed that the gills were the primary site for regulating alkalinity balance, while the kidneys played a secondary supportive role (Takvam et al., 2023). However, recent studies have

shown that the kidneys are crucial in regulating and eliminating acid-base equivalents and ammonia in fish, processes that are essential for adaptation to alkalinity stress (Tresguerres et al., 2020; Weiner and Verlander, 2011).

Current research on *S. argus* has primarily focused on saline stress, while the effects and response mechanisms of acute alkalinity stress on the kidneys of *S. argus* remain poorly understood (Yu et al., 2021; Pan et al., 2023; Zhong et al., 2021; Anney and Antony, 1988). This study investigates the histological changes in kidney tissue under acute alkalinity stress. As alkalinity stress progressed, several pathological changes were observed in the kidneys, including enlarged nuclei, increased intercellular spaces, vacuolization, and deformation and necrosis of glomeruli and renal tubules. These changes suggest that acute alkaline treatment cause substantial histological damage to the kidneys, impairing normal kidney functions. Similar histological changes have been reported in other fish species, such as *A. tarichi* (Oğuz, 2015), *C. auratus* (Ding et al., 2023; Lu et al., 2022) and bulatmai barbel (*Luciobarbus Capito*) (Zhang et al., 2021). The findings highlight the physiological challenges that fish face in alkaline environments, which can lead to significant kidney damage and impaired renal function. Previous research has indicated that enlargement of cell nuclei and the widening of intercellular spaces are indicative of significant cellular stress and damage (Rashid et al., 2023; Ho and Shirakawa, 2022). The vacuolization and necrosis observed in the glomeruli and renal tubules further highlight the extent of tissue damage, which may compromise the kidneys' ability to effectively filter and eliminate waste products (Gaut and Liapis, 2021). Moreover, the diminished antioxidant capacity observed in the kidneys under acute stress suggests an increase in oxidative stress. This disruption of the redox homeostasis within the body can result in further cellular damage and inflammation, exacerbating the negative effects on kidney function (McWilliam et al., 2021).

4.2. Acute alkaline treatment increases oxidative stress and disrupts the oxidation balance

In addition to histological changes, biochemical indicators can also reflect the effects of alkalinity stress on fish (Corbel, 1975). ALP and ACP are key enzymes in fish kidneys, with ALP playing a role in metabolic regulation, mineral absorption, and kidney function, while ACP is involved in intracellular digestion and tissue repair. Both enzymes are vital for maintaining normal kidney function and overall health in fish (Bedekar and Kv, 2022; Lallès, 2020). Elevated activities of ALP and ACP often indicate cellular damage or changes in metabolic activity,



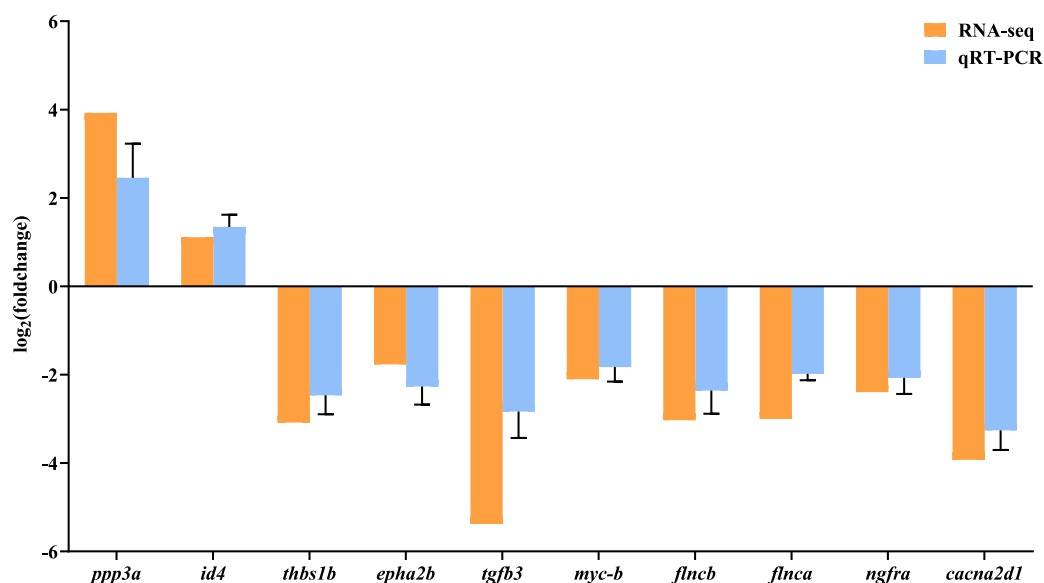


Fig. 7. Comparison of gene expression levels by RNA-seq and qRT-PCR of *S. argus* kidney in enrichment pathways.

reflecting the kidney's response to environmental stress (Makris et al., 2023; Mir et al., 2016; Zhu et al., 2024). In this study, significant differences in ALP and ACP activities were observed after acute alkalinity stress. As stress progressed, both enzyme activities increased, consistent with previous studies in Nile tilapia (*Oreochromis niloticus*) (Atencio et al., 2008). These findings suggest that acute alkalinity stress caused damage to kidney cells, altering the fish's immune and metabolic balance. Alkalinity stress is known to increase oxidative stress in fish kidneys, which may activate the expression of ALP and ACP (Liu et al., 2023). The elevated enzyme activities not only reflect the extent of kidney damage but are also linked to the overall immune response and metabolic changes in the fish (Liu et al., 2023). The increased ALP and ACP activities after acute alkalinity stress may result from metabolic adjustments and repair mechanisms initiated by the kidneys in response to stress, but these adaptive responses may also reduce the fish's ability to tolerate further stress.

Maintaining homeostasis in fish requires a delicate balance between oxidation and antioxidation (Duan et al., 2024). The equilibrium between excessive reactive oxygen species (ROS) production and antioxidant defenses is crucial for cellular homeostasis (Jomova et al., 2023). Disruption of this balance leads to oxidative stress, which can cause cellular damage and impair physiological functions. Studies have shown that fish exposed to alkalinity stress exhibit increased levels of ROS and MDA, indicating heightened oxidative stress. This stress response can impair kidney function, as the kidneys play a critical role in osmoregulation and detoxification (Ding et al., 2023; Menon et al., 2023). Three key antioxidant enzymes - CAT, SOD, and GSH-Px - work together to protect cells from peroxide damage and maintain normal physiological functions (Klein et al., 2017; Chu et al., 2023). In this study, MDA and GSH-Px levels in the kidneys increased over time, while SOD and CAT levels decreased, indicating a shift in the redox homeostasis after acute alkalinity stress. Elevated MDA activity suggests increased oxidative stress and tissue damage, while the reduced antioxidant capacity may impair fish health and kidney function (Song et al., 2023). The concurrent decrease in SOD and CAT activities, along with an increase in GSH-Px activity, reflects the dynamic regulation of the antioxidant defense system in *S. argus* under acute alkalinity stress. Under normal physiological conditions, SOD converts superoxide radicals (O_2^-) into hydrogen peroxide (H_2O_2), which is further broken down into water and oxygen by CAT (Ding et al., 2023; Olsvik et al., 2005). However, during the early phase of alkalinity stress, excessive ROS may cause direct oxidative damage to these enzymes, disrupting their structure and

leading to a rapid decline in activity (Ding et al., 2023). This impairment reduces the clearance efficiency of ROS, resulting in the accumulation of O_2^- and H_2O_2 and aggravating oxidative stress. In contrast, the up-regulation of GSH-Px activity appears to be a compensatory response. GSH-Px utilizes glutathione to reduce H_2O_2 and lipid peroxides, thereby partially mitigating oxidative damage in the context of reduced SOD and CAT function (Zhou et al., 2024a, 2024b). However, this secondary defense mechanism is insufficient to fully counterbalance the primary antioxidant impairment, eventually contributing to redox imbalance, kidney damage, and disrupted osmotic regulation. Additionally, the transient recovery of SOD activity observed at 72 h, followed by a subsequent decline, suggests a stage-specific adaptive response (Menon et al., 2023). Initially, SOD function is suppressed due to ROS overload; then, protective cellular mechanisms, such as transcriptional upregulation and enhanced enzyme synthesis, temporarily restore its activity. Nevertheless, persistent oxidative stress ultimately overwhelms these compensatory responses, leading to renewed enzymatic decline (Zhang et al., 2023a). Overall, these findings highlight the intricate interplay and temporal regulation of antioxidant enzymes in *S. argus* under alkalinity stress. The initial inhibition of SOD and CAT, together with a compensatory rise in GSH-Px activity, contributes to systemic antioxidant imbalance and subsequent physiological damage.

4.3. Acute alkaline treatment alters the mechanism and degree of response to alkalinity stress

Transcriptome analysis is a powerful tool for studying the physiological responses of organisms to environmental changes, offering critical insights into various biological processes in fish (Fang et al., 2022; Su et al., 2020). In stress experiments, transcriptome analysis can reveal alterations in gene expression, providing valuable information on the physiological and molecular mechanisms underlying stress responses (Liu et al., 2022). Previous studies on species like Japanese seaperch (*Lateolabrax japonicus*) and *C. auratus* have shown that changes in alkalinity can lead to significant fluctuations in gene expression (Zhang et al., 2023b; Ding et al., 2023). In this study, we utilized transcriptome analysis to explore the key molecular mechanisms of kidney response to acute alkalinity stress in *S. argus*. Our findings indicate that a total of 247 genes were differentially expressed after treatment, with 109 genes up-regulated and 138 genes down-regulated. Heatmap analysis further revealed distinct expression patterns after acute alkalinity stress. These changes in gene expression indicate that acute alkalinity stress induces

significant physiological adjustments in the kidneys of *S. argus*.

GO functional analysis revealed that alkalinity stress affects biological processes, cellular components, and molecular functions in the kidney. In our study, DEGs were significantly enriched in pathways related to immune response, immune system processes, cell adhesion, and protein folding. KEGG analysis identified 13 significantly enriched pathways, with the MAPK and TGF- β pathways being the most prominent, involving 13 DEGs. These findings suggest that alkalinity stress impacts kidney function through multiple biological processes and pathways (Ding et al., 2023; Wang et al., 2024). The significant enrichment of DEGs in immune-related pathways indicates that alkalinity stress may trigger an immune response in the kidney, potentially serving as a protective mechanism against cellular damage (Ding et al., 2023). Additionally, the involvement of cell adhesion and protein folding pathways suggests that the structural and functional integrity of kidney cells is compromised under stress (Samatova et al., 2024). The differential expression of DEGs in the MAPK and TGF- β pathways highlights their crucial roles in mediating the stress response.

The MAPK and TGF- β signaling pathways play indispensable roles in fish kidney function, especially under stress conditions such as alkalinity stress (Kajdaniuk et al., 2024; Dorry et al., 2024). In fish kidneys, the MAPK signaling pathway helps regulate cell proliferation and differentiation, ensuring and maintaining the basic functions of renal cells (Ihmann-Hella et al., 2014). The TGF- β signaling pathway is crucial in the repair process following injury or stress, helping to control inflammatory responses and prevent tissue damage (Giarratana et al., 2024). However, excessive activation of TGF- β signaling can lead to fibrosis, which is detrimental to kidney function (Giarratana et al., 2024). There is significant crosstalk between the MAPK and TGF- β signaling pathways, which is vital for balancing cell survival, proliferation, and differentiation under stress conditions. Both pathways influence the expression of genes involved in stress response and repair mechanisms and regulate enzymes like ALP and ACP, which are important for phosphate metabolism and lysosomal function (Kajdaniuk et al., 2024; Dorry et al., 2024). In this study, the expression of *ppp3ca* was up-regulated in the MAPK pathway. Previous research has shown that *ppp3ca* plays a critical role in the fish's response to alkalinity stress by maintaining osmotic balance through regulation of ion channels and transporters, modulating antioxidant enzymes to reduce oxidative stress, and influencing cell signaling and gene expression. These functions help enhance fish tolerance to alkaline environments (Wang et al., 2024; Su et al., 2020). The up-regulation of *ppp3ca* expression suggests improved osmotic regulation and antioxidant capacity, as well as the activation of adaptive responses to enhance survival in stressful environments (Su et al., 2020). Conversely, the expression of genes that promote fish growth and development and positively regulate the MAPK pathway, including *cacna2d1*, *ngfra*, *epha2b*, *tgfb3*, *flnca*, *flncb*, and *myc-b*, was significantly down-regulated. This down-regulation highlights the complex interaction between environmental stressors and genetic responses in fish, suggesting an inhibitory effect on the MAPK pathway. Such changes may reduce cellular proliferation and differentiation, potentially impairing overall fish growth and development under stress conditions (Xu et al., 2013a, 2013b; Shang et al., 2024; Zhao et al., 2023).

Significant differences of DEGs were also observed in the TGF- β pathway after acute alkalinity stress. The gene *id4*, which plays a role in regulating the cell cycle and maintaining cellular homeostasis, was found to be up-regulated (Roschger and Cabrele, 2017). In contrast, the genes *ths1b*, *tgfb3*, and *myc-b*, which are associated with stress adaptation in fish, exhibited down-regulation (García-Trevijano et al., 2013). These findings suggest that fish subjected to alkalinity stress are attempting to enhance cellular stability and regulate cell proliferation (Ding et al., 2023; Wang et al., 2024). This response is crucial for preventing uncontrolled cell growth and ensuring normal cellular function under stress conditions (Jiang et al., 2022). However, the down-regulation of *ths1b*, *tgfb3*, and *myc-b* genes indicates that stress

adaptation pathways may be inhibited, suggesting that the fish are experiencing stress levels beyond their typical adaptive capacity (Ding et al., 2023). This down-regulation could also reflect a shift in energy allocation, with the fish prioritizing immediate survival over long-term adaptation (Su et al., 2020; Harshini et al., 2023).

These findings indicate that alkalinity stress impacts the immune response and cellular function in *S. argus*. In response, *S. argus* enhance osmoregulation and antioxidant capacity to adapt to the high-stress environment. Concurrently, they reduce energy consumption by inhibiting cell proliferation and differentiation. This suppression of stress adaptation pathways indicates a strategic shift in energy allocation, prioritizing immediate survival over long-term adaptation.

To better elucidate the molecular adaptive mechanisms underlying acute alkalinity stress in *S. argus*, we propose the following hypothetical model (Fig. 8). Under high-alkalinity conditions, specialized receptors detect the altered external environment and trigger the closure of ion channels to limit ion influx, aiming to mitigate ion imbalance. This ion dysregulation leads to an increase in ROS, disrupting cellular oxidative homeostasis. As redox imbalance develops, key downstream signaling pathways, particularly MAPK and TGF- β , are activated. These pathways initiate oxidative stress responses and immune repair mechanisms, while also enhancing antioxidant capacity through the modulation of gene expression and suppressing cell proliferation and differentiation to conserve energy for immune repair. Despite these adaptive responses, the regulatory capacity is limited. When ROS levels exceed a threshold, these mechanisms fail, resulting in immune dysfunction. The ensuing antioxidant enzyme dysregulation exacerbates oxidative stress, culminating in significant tissue damage.

5. Conclusion

In summary, acute alkalinity stress induces significant histological damage to the kidneys of *S. argus*, including nuclear enlargement, vacuolization, and necrosis of renal tubules and glomeruli. Key enzyme activities, such as SOD, CAT, GSH-Px, ACP, and ALP, are notably altered under stress, indicating oxidative stress and changes in metabolic and immune functions. Transcriptome analysis identified 247 DEGs, with significant enrichment in immune response, cell adhesion, and protein folding pathways. The MAPK and TGF- β signaling pathways play crucial roles in the stress response, with up-regulation of genes like *ppp3ca* enhancing osmotic regulation and antioxidant capacity, while genes related to growth and development are down-regulated. These findings improve our understanding of the molecular mechanisms underlying the stress response in *S. argus* under alkalinity conditions and provide a scientific basis for exploring the tolerance and adaptive strategies of *S. argus* in dealing with such stress.

Ethics approval

The Guangdong Ocean University Animal Research and Ethics Committee authorized each and every experimental procedure used in this work (NIH Pub. No.85–23, amended 1996). There are no protected or endangered species included in this study. Every attempt was made to reduce the animals' suffering. Every experiment was carried out in compliance with all applicable laws and rules.

CRedit authorship contribution statement

Lingwei Kong: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Adili Abudu:** Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Lei Yang:** Investigation, Formal analysis, Data curation. **Minghui Ye:** Investigation, Formal analysis, Data curation. **Yulei Zhang:** Data curation. **Yucong Hong:** Funding acquisition. **Huapu Chen:** Writing – review & editing, Data curation. **Guangli Li:** Writing – review & editing, Funding acquisition, Data curation. **Gang**

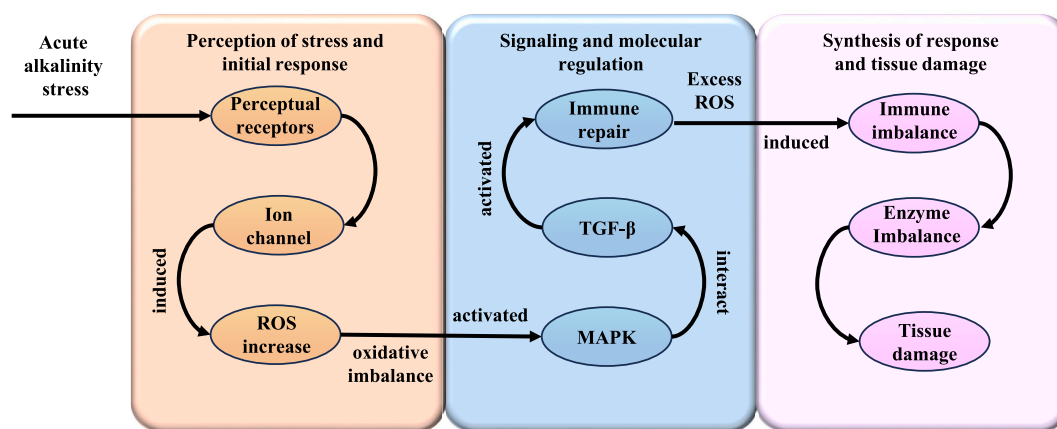


Fig. 8. Mechanism diagram of *S. argus* response to acute alkalinity stress.

Shi: Writing – original draft, Data curation. **Changxu Tian:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.aquaculture.2025.742601>.

References

- Afzal, M., Hindawi, S., Alghamdi, S., Migdadi, H., Khan, M., Hasnain, M., Sohaib, M., 2023. Potential breeding strategies for improving salt tolerance in crop plants. *J. Plant Growth Regul.* 42 (6), 3365–3387. <https://doi.org/10.1007/s00344-022-10797-w>.
- Akku, M., Sar, M., Ekmekçi, F., Yoğurtcuoğlu, B., 2021. The discovery of a microbialite-associated freshwater fish in the world's largest saline soda lake, Lake Van (Turkey). *Zoo. Evol.* 97 (1). <https://doi.org/10.3897/zse.97.62120>.
- Albecker, M., McCoy, M., 2017. Adaptive responses to salinity stress across multiple life stages in anuran amphibians. *Front. Zool.* 14, 1–16. <https://doi.org/10.1186/s12983-017-0222-0>.
- Anders, S., Huber, W., 2010. Differential expression analysis for sequence count data. *Nat. Preced.* 11 (10), R106. <https://doi.org/10.1038/npre.2010.4282.1>.
- Anney, M.K., Antony, A., 1988. Studies on some Aspects of Biology of Two Estuarine Fishes *Megalops Cyprinoides* and *Scatophagus Argus*. Doctoral dissertation, Cochran University Of Science And Technology. <https://api.semanticscholar.org/CorpusID:86594897>.
- Anufrieva, E., 2018. How can saline and hypersaline lakes contribute to aquaculture development? A review. *J. Oceanol. Limnol.* 36 (6), 2002–2009. <https://doi.org/10.1007/s00343-018-7306-3>.
- Atencio, L., Moreno, I., Prieto, A., Moyano, R., Molina, A., Cameán, A., 2008. Acute effects of microcystins MC-LR and MC-RR on acid and alkaline phosphatase activities and pathological changes in intraperitoneally exposed tilapia fish (*Oreochromis* sp.). *Toxicol. Pathol.* 36 (3), 449–458. <https://doi.org/10.1177/0192623308315356>.
- Barry, T., Fast, A., 1992. Biology of the spotted scat (*Scatophagus argus*) in the Philippines. *Asian Fish. Sci.* 5 (1992), 163–179. <https://doi.org/10.33997/j.afs.1992.5.2.003>.
- Bedekar, M.K., Kv, R., 2022. Overview of fish immune system. *Fish. Immun. Syst. Vacc.* 11–16. https://doi.org/10.1007/978-981-19-1268-9_1.
- Chang, Y., Liang, L., 2021. Advances of research of physiological and molecular mechanisms related to alkali-saline adaptation for fish species inhabiting alkali-saline water. *J. Fish. China* (In Chinese) 045 (005), 798–812. <https://doi.org/10.11964/jfc.20190311681>.
- Chen, J., Cai, B., Tian, C., Jiang, D., Shi, H., Huang, Y., Zhu, C., Li, G., Deng, S., 2023. RNA sequencing (RNA-Seq) analysis reveals liver lipid metabolism divergent adaptive response to low-and high-salinity stress in spotted scat (*Scatophagus argus*). *Animals* 13 (9), 1503. <https://doi.org/10.3390/ani13091503>.
- Chu, P., Luo, S., Wang, H., Zhang, K., Wen, X., Yin, S., Wang, T., 2023. Interactive effects of temperature and salinity on the apoptosis, antioxidant enzymes, and mapk signaling pathway of juvenile pufferfish (*Takifugu fasciatus*). *Aquac. Rep.* 29, 101483. <https://doi.org/10.1016/j.aqrep.2023.101483>.
- Corbel, M., 1975. The immune response in fish: a review. *J. Fish Biol.* 7 (4), 539–563. <https://doi.org/10.1111/j.1095-8649.1975.tb04630.x>.
- Ding, L., Liu, Y., Wei, X., Geng, C., Liu, W., Han, L., Sun, Y., 2023. Effects of saline-alkaline stress on metabolome, biochemical parameters, and histopathology in the kidney of crucian carp (*Carassius auratus*). *Metabolites* 13 (2), 159. <https://doi.org/10.3390/metabo13020159>.
- Dorry, S., Perla, S., Bennett, A., 2024. MAPK Phosphatase-5 is required for TGF-β signaling through a JNK-dependent pathway. *BioRxiv*. <https://doi.org/10.1101/2024.06.27.600976>.
- Duan, C., Tian, C., Guan, Y., Xu, H., Yang, L., Chen, Y., Liu, Y., Shen, Y., Zhang, Y., Cao, S., Huang, Y., Li, G., 2024. Long-term thermal stress induces hepatic injury and alters the thermotolerance response in Hong Kong catfish (*Clarias fuscus*). *Aquaculture* 590, 741041. <https://doi.org/10.1016/j.aquaculture.2024.741041>.
- Fahay, M., 2007. Early stages of fishes in the Western North Atlantic Ocean. *J. Northwest Atl. Fish. Sci.* 1–1696. <https://journal.nafo.int/Portals/0/2007-2008>.
- Fan, Z., Wu, D., Zhang, Y., Li, J., Xu, Q., Wang, L., 2021. Carbonate alkalinity and dietary protein levels affected growth performance, intestinal immune responses and intestinal microflora in Songpu mirror carp (*Cyprinus carpio* Songpu). *Aquaculture* 545, 737135. <https://doi.org/10.1016/j.aquaculture.2021.737135>.
- Fang, H., Yang, Y., Wu, X., Zheng, S., Song, Y., Zhang, J., Chang, M., 2022. Effects and molecular regulation mechanisms of salinity stress on the health and disease resistance of grass carp. *Front. Immunol.* 13, 917497. <https://doi.org/10.3389/fimmu.2022.917497>.
- FAO, 2024. The state of world fisheries and aquaculture. Rome, pp. 1–7. <https://doi.org/10.4060/cd0683en>.
- Gandhi, V., 2002. Studies on the food and feeding habits of cultivable butterfish *Scatophagus argus* (Cuv. And Val.). *J. Mar. Biol. Assoc. India* 44 (1 & 2), 115–121. <http://eprints.cmfri.org.in/id/eprint/1988>.
- Garber, M., Grabherr, M., Guttman, M., Trapnell, C., 2011. Computational methods for transcriptome annotation and quantification using RNA-seq. *Nat. Methods* 8 (6), 469–477. <https://doi.org/10.1038/nmeth.1613>.
- García-Trevijano, E., Torres, L., Zaragoza, R., Viña, J., 2013. The role of Id2 in the regulation of chromatin structure and gene expression. *Chromatin Remodelling* 91. <https://doi.org/10.5772/54969>.
- Gaut, J., Liapi, H., 2021. Acute kidney injury pathology and pathophysiology: a retrospective review. *Clin. Kidney J.* 14 (2), 526–536. <https://doi.org/10.1093/ckj/sfaa142>.

- Giarratana, A., Prendergast, C., Salvatore, M., Capaccione, K., 2024. TGF- β signaling: critical nexus of fibrogenesis and cancer. *J. Transl. Med.* 22 (1), 594. <https://doi.org/10.1186/s12967-024-05411-4>.
- Gupta, S., 2016. An overview on morphology, biology, and culture of spotted scat *Scatophagus argus* (Linnaeus 1766). *Rev. Fish. Sci. Aquac.* 24 (2), 203–212. <https://doi.org/10.1080/23308249.2015.1119800>.
- Harshini, V., Shukla, N., Raval, I., Kumar, S., Shrivastava, V., Chaudhari, A., Patel, A., Joshi, C., 2023. Interplay of gene expression and regulators under salinity stress in gill of *Labeo rohita*. *BMC Genomics* 24 (1), 336. <https://doi.org/10.1186/s12864-023-09426-x>.
- Ho, H., Shirakawa, H., 2022. Oxidative stress and mitochondrial dysfunction in chronic kidney disease. *Cells* 12 (1), 88. <https://doi.org/10.3390/cells12010088>.
- Huang, Y., Mustapha, U.F., Huang, Y., Tian, C., Yang, W., Chen, H., Deng, S., Zhu, C., Jiang, D., Li, G., 2021. A chromosome-level genome assembly of the spotted scat (*Scatophagus argus*). *Genome Biol. Evol.* 13 (6), evab092. <https://doi.org/10.1093/gbe/evab092>.
- Ihermann-Hella, A., Lume, M., Miinalainen, I.J., Pirttiniemi, A., Gui, Y., Peränen, J., Charron, J., Saarma, M., Costantini, F., Kuure, S., 2014. Mitogen-activated protein kinase (MAPK) pathway regulates branching by remodeling epithelial cell adhesion. *PLoS Genet.* 10 (3), e1004193. <https://doi.org/10.1371/journal.pgen.1004193>.
- Jiang, Y., Yuan, C., Qi, M., Liu, Q., Hu, Z., 2022. The effect of salinity stress on enzyme activities, histology, and transcriptome of silver carp (*Hypophthalmichthys molitrix*). *Biology* 11 (11), 1580. <https://doi.org/10.3390/biology11111580>.
- Jomova, K., Raptova, R., Alomar, S., Alwasel, S., Nepovimova, E., Kuca, K., Valko, M., 2023. Reactive oxygen species, toxicity, oxidative stress, and antioxidants: chronic diseases and aging. *Arch. Toxicol.* 97 (10), 2499–2574. <https://doi.org/10.1007/s00204-023-03562-9>.
- Kajdaniuk, D., Hudy, D., Strzelczyk, J.K., Mlynarek, K., Slomian, S., Potyka, A., Szymonik, E., Strzelczyk, J., Foltyn, W., Kos-Kudla, B., Marek, B., 2024. Transforming growth factors β and their signaling pathway in renal cell carcinoma and peritumoral space - transcriptome analysis. *Clin. Transl. Oncol.* 26 (5), 1229–1239. <https://doi.org/10.1007/s12094-023-03350-y>.
- Kao, M., Ang, D., Pall, A., Struthers, A., 2010. Oxidative stress in renal dysfunction: mechanisms, clinical sequelae and therapeutic options. *J. Hum. Hypertens.* 24 (1), 1–8. <https://doi.org/10.1038/jhh.2009.70>.
- Klein, R., Borges, V., Rosa, C., Colares, E., Robaldo, R., Martinez, P., Bianchini, A., 2017. Effects of increasing temperature on antioxidant defense system and oxidative stress parameters in the Antarctic fish *Notothenia coriiceps* and *Notothenia rossii*. *J. Therm. Biol.* 68, 110–118. <https://doi.org/10.1016/j.jtherbio.2017.02.016>.
- Kültz, D., 2015. Physiological mechanisms used by fish to cope with salinity stress. *J. Exp. Biol.* 218 (12), 1907–1914. <https://doi.org/10.1242/jeb.118695>.
- Lallès, J., 2020. Intestinal alkaline phosphatase in the gastrointestinal tract of fish: biology, ontogeny, and environmental and nutritional modulation. *Rev. Aquac.* 12 (2), 555–581. <https://doi.org/10.1111/raq.12340>.
- Li, Y., Gao, P., Zhou, K., Yao, Z., Sun, Z., Qin, H., Lai, Q., 2024a. Effects of saline and alkaline stresses on the survival, growth, and physiological responses in juvenile mandarin fish (*Siniperca chuatsi*). *Aquaculture* 741143. <https://doi.org/10.1016/j.aquaculture.2024.741143>.
- Li, W., Wang, J., Li, J., Liu, P., Fei, F., Liu, B., Li, J., 2024b. The effect of astaxanthin on the alkalinity stress resistance of *Exopalaemon carinicauda*. *Sci. Total Environ.* 917, 170415. <https://doi.org/10.1016/j.scitotenv.2024.170415>.
- Liao, Y., Smyth, G., Shi, W., 2014. FeatureCounts: an efficient general purpose program for assigning sequence reads to genomic features. *Bioinformatics* 30 (7), 923–930. <https://doi.org/10.1093/bioinformatics/btt656>.
- Liu, J., Zhang, X., Sheng, J., 2022. Integrative analysis of the transcriptome and metabolome reveals the mechanism of saline-alkali stress tolerance in *Astragalus membranaceus* (Fisch) Bge var. *mongolicus* (Bge.) Hsiao. *Food Qual. Saf.* 6, fyac001. <https://doi.org/10.1093/fqsafe/fyac001>.
- Liu, Y., Yang, Y., Qin, R., Peng, Y., Huang, Y., Zhu, C., Li, G., Jiang, D., Shi, H., 2023. Divergent molecular responses of greater amberjack (*Seriola lalandi*) to acute salinity stress revealed by comparative transcriptome analysis. *Front. Mar. Sci.* 10, 1185015. <https://doi.org/10.3389/fmars.2023.1185015>.
- Lu, Y., Zhang, Y., Zhang, P., Liu, J., Wang, B., Bu, X., Wei, Q., Liu, S., Li, Y., 2022. Effects of dietary supplementation with *Bacillus subtilis* on immune, antioxidant, and histopathological parameters of *Carassius auratus gibelio* juveniles exposed to acute saline-alkaline conditions. *Aquac. Int.* 30 (5), 2295–2310. <https://doi.org/10.1007/s10499-022-00902-x>.
- Makris, K., Mousa, C., Cavalier, E., 2023. Alkaline phosphatases: biochemistry, functions, and measurement. *Calcif. Tissue Int.* 112 (2), 233–242. <https://doi.org/10.1007/s00223-022-01048-x>.
- McWilliam, S.J., Wright, R.D., Welsh, G.I., Tuffin, J., Budge, K.L., Swan, L., Wilm, T., Martinis, I., Littlewood, J., Oni, L., 2021. The complex interplay between kidney injury and inflammation. *Clin. Kidney J.* 14 (3), 780–788. <https://doi.org/10.1093/cjk/sfaa164>.
- Menon, S.V., Kumar, A., Middha, S.K., Paital, B., Mathur, S., Johnson, R., Asthana, M., 2023. Water physicochemical factors and oxidative stress physiology in fish, a review. *Front. Env. Sci.* 11, 1240813. <https://doi.org/10.3389/fenvs.2023.1240813>.
- Mir, G., Tharani, M., Hussain, A., Ahmad, Y., Rashid, A., 2016. Variations in acid phosphatase (ACP) and alkaline phosphatase (ALP) activities in liver and kidney of a freshwater fish *Labeo rohita* exposed to heavy metal concentrations. *European J. Pharm. Med. Res.* 3 (3), 398–401. <https://api.semanticscholar.org/CorpusID:90034479>.
- Mu, X., Su, M., Gui, L., Liang, X., Zhang, P., Hu, P., Liu, Z., Zhang, J., 2015. Comparative renal gene expression in response to abrupt hypoosmotic shock in spotted scat (*Scatophagus argus*). *Gen. Comp. Endocrinol.* 215, 25–35. <https://doi.org/10.1016/j.ygcen.2014.09.017>.
- Naylor, R., Hardy, R., Buschmann, A., Rush, S., Cao, L., Klinger, D., Little, D., Lubchenco, J., Shumway, S., Troell, M., 2021. A 20-year retrospective review of global aquaculture. *Nature* 591 (7851), 551–563. <https://doi.org/10.1038/s41586-021-03308-6>.
- Oğuz, A., 2015. A histological study of the kidney structure of Van fish (*Alburnus tarichi*) acclimated to highly alkaline water and freshwater. *Mar. Freshw. Behav. Physiol.* 48 (2), 135–144. <https://doi.org/10.1080/10236244.2015.1004838>.
- Olsvik, P., Kristensen, T., Waagbø, R., Rosseland, B., Tollefsen, K., Baeverfjord, G., Berntsen, M., 2005. mRNA expression of antioxidant enzymes (SOD, CAT and GSH-Px) and lipid peroxidative stress in liver of Atlantic salmon (*Salmo salar*) exposed to hyperoxic water during smoltification. *Comp. Biochem. Physiol. C Toxicol. Pharmacol.* 141 (3), 314–323. <https://doi.org/10.1016/j.cbpc.2005.07.009>.
- Overton, J., Bayley, M., Paulsen, H., Wang, T., 2008. Salinity tolerance of cultured Eurasian perch, *Perca fluviatilis* L.: effects on growth and on survival as a function of temperature. *Aquaculture* 277 (3–4), 282–286. <https://doi.org/10.1016/j.aquaculture.2008.02.029>.
- Pan, Q., Zhai, Y., Cai, B., Jiang, D., Shi, H., Huang, Y., Li, G., Deng, S., 2023. Ovarian transcriptome analysis in *Scatophagus argus* under chronic salinity stress. *J. Fish. Sci. China* 30 (3), 324–333. <https://doi.org/10.12264/JFSC2022-0378>.
- Pang, C., Ip, Y., Chew, S., 2021. Effects of seawater acclimation on two Na⁺/K⁺-ATPase α -subunit isoforms in the gills of the marble goby *Oxyeleotris marmorat*. *Comp. Biochem. Physiol.* 253, 110853. <https://doi.org/10.1016/j.cbpa.2020.110853>.
- Partelow, S., Nagel, B., Gentry, R., Gephart, J., Rocha, J., 2025. Archetypes of aquaculture development across 150 countries. *Aquaculture* 595, 741484. <https://doi.org/10.1016/j.aquaculture.2024.741484>.
- Petitjean, Q., Jean, S., Gandar, A., Côte, J., Laffaille, P., Jacquin, L., 2019. Stress responses in fish: from molecular to evolutionary processes. *Sci. Total Environ.* 684, 371–380. <https://doi.org/10.1016/j.scitotenv.2019.05.357>.
- Qin, Z., Ge, Q., Wang, J., Li, M., Liu, P., Li, J., Li, J., 2021. Comparative transcriptomic and proteomic analysis of *Exopalaemon carinicauda* in response to alkalinity stress. *Front. Mar. Sci.* 8, 759923. <https://doi.org/10.3389/fmars.2021.759923>.
- Rashid, H., Jali, A., Akhter, M., Abdi, S., 2023. Molecular mechanisms of oxidative stress in acute kidney injury: targeting the loci by resveratrol. *Int. J. Mol. Sci.* 25 (1), 3. <https://doi.org/10.3390/ijms25010003>.
- Roschger, C., Cabrele, C., 2017. The id-protein family in developmental and cancer-associated pathways. *Cell Commun. Signal* 15, 1–26. <https://doi.org/10.1186/s12964-016-0161-y>.
- Samatova, E., Komar, A., Rodnina, M., 2024. How the ribosome shapes cotranslational protein folding. *Curr. Opin. Struct. Biol.* 84, 102740. <https://doi.org/10.1016/j.sbi.2023.102740>.
- Shang, X., Geng, L., Wei, H., Liu, T., Che, X., Li, W., Liu, Y., Shi, X., Li, J., Teng, X., Xu, W., 2024. Analysis revealed the molecular mechanism of oxidative stress-autophagy-induced liver injury caused by high alkalinity: integrated whole hepatic transcriptome and metabolome. *Front. Immunol.* 15, 1431224. <https://doi.org/10.3389/fimmu.2024.1431224>.
- Sivan, G., 2011. Food, feeding habits and biochemical composition of *Scatophagus argus*. *Turk. J. Fish. Aquat. Sci.* 11 (4). https://doi.org/10.4194/1303-2712-v11_4_14.
- Smyth, K., Elliott, M., 2016. Effects of changing salinity on the ecology of the marine environment. Stressors in the marine environment: physiological and ecological responses. *Societal Implicat.* 161–174. <https://doi.org/10.1093/acprof:oso/9780198718826.001.0001>.
- Song, C., Sun, C., Liu, B., Xu, P., 2023. Oxidative stress in aquatic organisms. *Antioxidant* 12 (6), 1223. <https://doi.org/10.3390/antiox12061223>.
- Su, H., Ma, D., Zhu, H., Liu, Z., Gao, F., 2020. Transcriptomic response to three osmotic stresses in gills of hybrid tilapia (*Oreochromis mossambicus* female $\times$ *O. urolepis hornorum* male). *BMC Genomics* 21, 1–20. <https://doi.org/10.1186/s12864-020-6512-5>.
- Takvam, M., Wood, C., Kryvi, H., Nilsen, T., 2023. Role of the kidneys in acid-base regulation and ammonia excretion in freshwater and seawater fish: implications for nephrocalcinosis. *Front. Physiol.* 14, 1226068. <https://doi.org/10.3389/fphys.2023.1226068>.
- Tong, C., Zhang, C., Zhang, R., Zhao, K., 2015. Transcriptome profiling analysis of naked carp (*Gymnocypris przewalskii*) provides insights into the immune-related genes in highland fish. *Fish Shellfish Immunol.* 46 (2), 366–377. <https://doi.org/10.1016/j.fsi.2015.06.025>.
- Tresguerres, M., Clifford, A., Harter, T., Roa, J., Thies, A., Yee, D., Brauner, C., 2020. Evolutionary links between intra-and extracellular acid-base regulation in fish and other aquatic animals. *J. Exp. Zool.* 333 (6), 449–465. <https://doi.org/10.1002/jez.2367>.
- United Nations, 2019. *World population prospects 2019*. NewYork.
- Wang, Y.S., Gonzalez, R.J., Patrick, M.L., Grosell, M., Zhang, C., Feng, Q., Du, J., Walsh, P., Wood, C.M., 2003. Unusual physiology of scale-less carp, *Gymnocypris przewalskii*, in Lake Qinghai: a high altitude alkaline saline lake. *Comp. Biochem. Physiol.* 134 (2), 409–421. [https://doi.org/10.1016/S1095-6433\(02\)00317-3](https://doi.org/10.1016/S1095-6433(02)00317-3).
- Wang, Z., Hu, J., Mei, Z., Zhang, Y., Liu, Q., Yang, D., 2023. Mannan-oligosaccharide induces trained immunity activation and alleviates pathological liver injury in turbot (*Scophthalmus maximus*). *Aquaculture* 578, 740097. <https://doi.org/10.1016/j.aquaculture.2023.740097>.
- Wang, M., Yan, Y., Liu, W., Fan, J., Li, E., Chen, L., Wang, X., 2024. Proline metabolism is essential for alkaline adaptation of Nile tilapia (*Oreochromis niloticus*). *J. Anim. Sci. Biotechnol.* 15 (1), 142. <https://doi.org/10.1186/s40104-024-01100-w>.
- Wang, M., Li, E., Huang, Y., Liu, W., Wang, S., Li, W., Chen, L., Wang, X., 2025. Dietary supplementation with glutamate enhanced antioxidant capacity, ammonia detoxification and ion regulation ability in Nile tilapia (*Oreochromis niloticus*) exposed to acute alkalinity stress. *Aquaculture* 594, 741360. <https://doi.org/10.1016/j.aquaculture.2024.741360>.

- Weiner, I., Verlander, J., 2011. Role of NH_3 and NH_4^+ transporters in renal acid-base transport. *Am. J. Physiol. Ren. Physiol.* 300 (1), 11–23. <https://doi.org/10.1152/ajprenal.00554.2010>.
- Wen, J., Chen, S., Xu, W., Zheng, G., Zou, S., 2023. Effects of high NaHCO_3 alkalinity on growth, tissue structure, digestive enzyme activity, and gut microflora of grass carp juvenile. *Environ. Sci. Pollut. Res.* 30 (36), 85223–85236. <https://doi.org/10.1007/s11356-023-28083-x>.
- Wilkie, M., Wood, C., 1996. The adaptations of fish to extremely alkaline environments. *Comp. Biochem. Physiol.* 113 (4), 665–673. [https://doi.org/10.1016/0305-0491\(95\)02092-6](https://doi.org/10.1016/0305-0491(95)02092-6).
- Wright, P., Iwama, G., Wood, C., 1993. Ammonia and urea excretion in Lahontan cutthroat trout (*Oncorhynchus clarki henshawi*) adapted to the highly alkaline pyramid Lake (pH 9.4). *J. Exp. Biol.* 175 (1), 153–172. <https://doi.org/10.1242/jeb.175.1.153>.
- Xu, J., Ji, P., Wang, B., Zhao, L., Wang, J., Zhao, Z., Li, J., Xu, P., Sun, X., 2013a. Transcriptome sequencing and analysis of wild Amur ide (*Leuciscus waleckii*) inhabiting an extreme alkaline-saline lake reveals insights into stress adaptation. *PLoS One* 8 (4), e59703. <https://doi.org/10.1371/journal.pone.0059703>.
- Xu, J., Li, Q., Xu, L., Wang, S., Jiang, Y., Zhao, Z., Zhang, Y., Li, J., Dong, C., Xu, P., Sun, X., 2013b. Gene expression changes leading extreme alkaline tolerance in Amur ide (*Leuciscus waleckii*) inhabiting soda lake. *BMC Genomics* 14, 1–11. <https://doi.org/10.1186/1471-2164-14-682>.
- Yao, Z.L., Lai, Q.F., Zhou, K., Rizalita, R.E., Wang, H., 2010. Developmental biology of medaka fish (*Oryzias latipes*) exposed to alkalinity stress. *J. Appl. Ichthyol.* 26 (3), 397–402. <https://doi.org/10.1111/j.1439-0426.2009.01360.x>.
- Yu, G., Wang, L., Han, Y., He, Q., 2012. ClusterProfiler: an R package for comparing biological themes among gene clusters. *OMICS. J. Integr. Biol.* 16 (5), 284–287. <https://doi.org/10.1089/omi.2011.0118>.
- Yu, Y., Liu, J., Feng, G., Zhao, F., Sun, X., Wang, M., Zou, X., Huang, X., Li, Q., Sun, Y., Zheng, W., 2021. Effect of abrupt salinity change on morphology and structure of mitochondria-rich cells in *Scatophagus argus*. *South China Fisheries Science* (In Chinese) 17 (2), 60–69. <https://doi.org/10.12131/20200214>.
- Zhang, Y., Yang, J., Geng, L., Wang, Y., Li, C., 2021. Effect of NaHCO_3 alkalinity on oxidative stress of *Luciobarbus capito*. *Period. Ocean. Univ. China* (In Chinese) 51, 32–39. <https://doi.org/10.16441/j.cnki.hdxh.20200223>.
- Zhang, R., Shi, X., Liu, Z., Sun, J., Sun, T., Lei, M., 2023a. Histological, physiological and transcriptomic analysis reveal the acute alkalinity stress of the gill and hepatopancreas of *Litopenaeus vannamei*. *Mar. Biotechnol.* 25 (4), 588–602. <https://doi.org/10.1007/s10126-023-10228-1>.
- Zhang, Y., Wen, H., Liu, Y., Qi, X., Sun, D., Zhang, C., Zhang, K., Zhang, M., Li, J., Li, Y., 2023b. Gill histological and transcriptomic analysis provides insights into the response of spotted sea bass (*Lateolabrax maculatus*) to alkalinity stress. *Aquaculture* 563, 738945. <https://doi.org/10.1016/j.aquaculture.2022.738945>.
- Zhang, R., Zhao, Z., Li, M., Luo, L., Wang, S., Guo, K., Xu, W., 2023c. Effects of saline-alkali stress on the tissue structure, antioxidation, immunocompetence and metabolomics of Eriocheir sinensis. *Sci. Total Environ.* 871, 162109. <https://doi.org/10.1016/j.scitotenv.2023.162109>.
- Zhao, Y., Zhang, C., Zhou, H., Song, L., Wang, J., Zhao, J., 2020. Transcriptome changes for Nile tilapia (*Oreochromis niloticus*) in response to alkalinity stress. *Comp. Biochem. Physiol.* 33, 100651. <https://doi.org/10.1016/j.cbd.2019.100651>.
- Zhao, Y., Li, S., Tang, S., Wang, Y., Yao, X., Xie, J., Zhao, J., 2023. Effects of chloride, sulfate, and bicarbonate stress on mortality rate, gill tissue morphology, and gene expression in mandarin fish (*Siniperca chuatsi*). *Environ. Sci. Pollut. Res.* 30 (44), 99440–99453. <https://doi.org/10.1007/s11356-023-29411-x>.
- Zhong, Y., Duan, Z., Su, M., Lin, Y., Zhang, J., 2021. Inflammatory responses associated with hyposaline stress in gill epithelial cells of the spotted scat *Scatophagus argus*. *Fish Shellfish Immunol.* 114, 142–151. <https://doi.org/10.1016/j.fsi.2021.04.023>.
- Zhou, H., Yao, T., Zhang, T., Sun, M., Ning, Z., Chen, Y., Mu, W., 2024a. Effects of chronic saline-alkalinity stress on gill, liver and intestinal histology, biochemical, and immune indexes in Amur minnow (*Phoxinus lagowskii*). *Aquaculture* 579, 740153. <https://doi.org/10.1016/j.aquaculture.2023.740153>.
- Zhou, F., Chang, M., Ruan, S., Huang, W., Sha, Z., Cai, B., Liu, Z., 2024b. Transcriptomic and histologic analyses preliminarily reveal the immune-metabolic response mechanism to saline-alkaline in large yellow croaker (*Larimichthys crocea*). *Comp. Biochem. Physiol.* 52, 101282. <https://doi.org/10.1016/j.cbd.2024.101282>.
- Zhou, B., Sui, R., Yu, L., Qi, D., Fu, S., Luo, Y., Qi, H., Tian, F., 2025. Transcriptomics and proteomics provide insights into the adaptative strategies of Tibetan naked carps (*Gymnocypris przewalskii*) to saline-alkaline variations. *BMC Genomics* 26, 162. <https://doi.org/10.1186/s12864-025-11336-z>.
- Zhu, X., Jin, Y., Zhang, X., Liu, Q., Tang, B., 2024. Transcriptome analysis reveals immune and antioxidant defense mechanisms in the Eriocheir japonica sinensis after exposure to Ammonia. *Animals* 14 (20), 2981. <https://doi.org/10.3390/ani14202981>.